Fiscal Year:	FY 2011	**Task Last Updated:**	FY 10/26/2010
PI Name:	Bhattacharjee, Subrata Ph.D.		
Project Title:	Residence Time Driven Flame Spread Over Solid Fuels		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	92182-0001	**Congressional District:**	53
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
Start Date:	01/01/2010	**End Date:**	12/31/2012
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	8	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	10	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Miller, Fletcher (San Diego State University) Paolini, Christopher (San Diego State University) Takahashi, Shuhei (Gifu University) Wakai, Kazunori (Gifu University)		
Grant/Contract No.:	NNX10AE03G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Flame spread over solid fuels in an opposed-flow environment has been investigated for over four decades for understanding the fundamental nature of hazardous fire spread. The appeal for this configuration stems from the fact that flame spread rate remains steady, even if the flame itself may grow in size. For practical fire safety issues, however, wind-assisted flame spread is more relevant. However, these two regimes have always been studied in isolation without much effort to establish a connection, even though the underlying mechanism of flame spread is the same in all regimes. Sitting between the two regimes are high-residence time flames, as found in a low-velocity or quiescent microgravity environment. Residence time is the time spent by an oxidizer in the combustion zone. Such flames, which are of interest on their own merit due to fire safety issues in spacecrafts, offer some unique characteristics because of the high residence time. Radiation becomes dominant and, based on previous space experiments and analysis, we contend that a vigorously spreading flame on Earth becomes self-extinguishing in a microgravity environment under certain conditions such as the fuel thickness being greater than a critical value. The proposed research uses a comprehensive approach-- a novel experimental set up and a theoretical framework based on scaling and numerical modeling-- to investigate flame spread driven by varying residence time, from blow-off extinction in an opposed-flow configuration through high residence time flame to blow-off extinction in a concurrent-flow configuration. At the heart of this proposal is a novel but simple experiment where the residence time of the oxidizer can be controlled and high residence time flames can be established for a long duration (compared to drop towers). As a proof of concept, we have constructed a flame tower at SDSU in which, after a sample is ignited, the sample holder, placed in an open moveable cart, can be traversed at any desired speed upward or downward, creating an external flow that can augment or mitigate the buoyancy-induced flow. Preliminary results show that we can control the residence time and create flames in different regimes, including a transition between a wind-aided and wind-opposed configuration. At Gifu University in Japan, we have been developing an interferometry based imaging system which we intend to enhance to capture the thermal footprint of a flame's leading edge. The leading edge is central to our understanding of mechanism of flame extinction. Further development of this technology will enable us to integrate diagnostics in future space based experiments and provide validation data to a comprehensive numerical model. The comprehensive model, to be built upon our existing two-dimensional model, will solve an unsteady, three-dimensional, Navier stokes equation with finite rate kinetics in the gas and solid phases and radiation in the gas phase. The software implementation will be object-oriented and utilize a new technology called Web Services that will decouple various sub-models and enhance parallel execution. The radiation model will also be refined by including the equilibrium composition of species for finding radiative properties in high residence-time flames. The new data generated by our proposed ground-based experiment will be the first of its kind to encompass flame spread across different regimes and will stimulate fundamental theoretical research to unify different regimes of flame spread. The comprehensive model, tested against available theory, data in literature, and data generated at SDSU and Gifu, will be applied to test the three hypotheses presented in this proposal regarding flame extinguishment in a microgravity environment. A successful outcome of this project will lead to a well thought out space-based experiment on the mechanism of flame extinction in a gravity free environment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research have three components: We are building a novel experimental facility called Flame Tower at SDSU, where ground based experiments on flame spread will be conducted; Theoretical and computational work that explores the similarity and differences between the mechanisms flame spread in a zero gravity space environment and on Earth; Design and propose a space based experiment to establish extinction mechanism of flames. The data that we plan to acquire in the Flame Tower will provide the research community with a comprehensive set of results for testing different theories of flame spread in a normal gravity environment. Moreover, by controlling the residence time, various regimes of flame spread, including the microgravity regime, can be explored in the Flame Tower. Our theoretical work predicts a fuel thickness beyond which steady flame spread is unsustainable in a gravity free environment. If we are successful in establishing a critical thickness, this will have a powerful impact on making fire resistant environment for humans in space.
Task Progress:	1. Progress in building the Flame Tower: The proposed flame tower is the center piece of our ground based activities. When completed, it will provide a regulated environment in terms of ambient pressure and oxygen concentration in a 8 m tall closed chamber. The sample holder assembly, mounted on a vertical rail, can be moved up or down at any desired velocity, thus, creating a known free-stream flow (concurrent or opposed) relative to the fuel sample. In this period, we have considered different design ideas and finally converged on a ladder like structure, which had several advantages over other ideas. Most importantly, it was very simple consisting of only three different pieces: vertical columns, horizontal braces, and skin sheets. Steel was selected as the working substance and a finite element analysis was carried out to ensure that the structure can withstand low internal pressure. The structure was fabricated at SDSU spanning two floors in the engineering building. We hope to construct the motion system for the sample holder and an automated sample carriage in the near future. Our goal is be able to acquire data under atmospheric condition by the next reporting period. 2. Progress in Building the Flame Tracker: This apparatus will allow us to measure instantaneous flame spread rate and monitor changes in flame shape or size in real time by moving the sample in opposite direction of flame spread so that the flame leading edge remains stationary. We have already built this apparatus in the laboratory and demonstrated that a spreading flame can be converted into a stationary flame. We are now working on miniaturizing the system so that it can be mounted in the flame tower and moved up and down at any desired velocity. 3. Scale Analysis: We have continued further development of scale analysis of the flame to produce formulas for flame shape and flame size for opposed-flow flame spread in the thermal regime. A paper was presented in the 33rd Symposium on Combustion on this work. 4. Computational Tools: We have developed a rich internet application to calculate equilibrium flame temperature for different kinds of fuels. We plan to use these web service based tool to develop a flame spread rate calculator.

	5. Computational Simulation: We have been using the existing numerical model to explore thickness effect on flame spread rate. We plan to expand this study to map the entire radiative extinction regime. 6. Space Experiment Concepts: We are working on a simplified space experiment that can establish the central thrust of our proposal - to establish that the separation between thermal and species field is responsible for flame extinction in the radiative microgravity regime. Two preliminary concepts are under development at this stage. In the next couple of months, we plan to have these concepts fleshed out and ready to be integrated with experiments from other PIs.
Bibliography Type:	Description: (Last Updated: 03/31/2026)
Articles in Peer-reviewed Journals	Bhattacharjee S, Takahashi S, Wakai K, Paolini C. "Correlating flame geometry in opposed-flow flame spread over thin fuels." Proceedings of the Combustion Institute. 2011;33(2):2465-72. https://doi.org/10.1016/j.proci.2010.06.053 , Jan-2011
Articles in Peer-reviewed Journals	Paolini C, Bhattacharjee S. "Solving chemical equilibrium problems online." Journal of Chemical Education. 2010 Apr;87(4):456. http://dx.doi.org/10.1021/ed800134r , Apr-2010
Papers from Meeting Proceedings	Patterson M, Paolini C, Bhattacharjee S. "Design and Implementation of a Rich Internet Application (RIA) for the Simulation of a Combustion Chamber." ASEE Annual Conference and Exposition, Louisville, KY, June 20-23, 2010. ASEE Annual Conference and Exposition Proceedings, 2010. http://soa.asee.org/paper/conference/paper-view.cfm?id=24720 , Jun-2010